

## UNAHA, Russian Federation

WMO: 311990

Lat: 55.033N

Lon: 126.800E

Elev: 1470

StdP: 13.93

Time Zone: 9.00 (E09)

Period: 82-94

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -45.4      | -41.1 | -49.6                         | 0.3 | -44.3 | -45.5 | 0.4 | -40.1 | 12.2                  | -5.9 | 11.1 | -3.0 | 0.0                   | N/A  | 0.671 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.8                   | 81.9            | 63.0 | 78.9 | 62.3 | 75.5 | 61.1 | 67.0                | 75.8 | 65.2 | 73.8 | 63.4 | 71.3 | 4.1                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.2                            | 94.9 | 70.3 | 62.3 | 88.8 | 67.9 | 60.5 | 83.2 | 66.7 | 32.5          | 75.5 | 31.1 | 73.6 | 29.7 | 71.1 | 77.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.8          | 9.7  | 7.9 | DB                         | -51.6 | 88.2               | 2.7 | 1.8                                                | -53.5 | 89.5       | -55.1 | 90.5       | -56.7 | 91.5       | -58.6 | 92.7 | (4) |
| (5)               |      |     | WB                         | -51.0 | 70.2               | 2.2 | 2.0                                                | -52.6 | 71.6       | -53.8 | 72.8       | -55.1 | 73.9       | -56.6 | 75.4 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 23.0  | -17.7 | -9.6  | 6.1   | 27.7  | 44.1  | 57.4  | 61.6  | 56.8  | 43.7  | 22.5  | -2.3  | -16.1 | (6)  |
| (7)  |                                                                   | DBStd       | 29.95 | 10.47 | 10.23 | 9.58  | 9.10  | 7.80  | 6.28  | 5.22  | 4.85  | 7.33  | 10.21 | 9.20  | 10.21 | (7)  |
| (8)  |                                                                   | HDD50       | 10714 | 2097  | 1667  | 1360  | 670   | 216   | 11    | 0     | 7     | 214   | 854   | 1570  | 2048  | (8)  |
| (9)  |                                                                   | HDD65       | 15358 | 2562  | 2087  | 1825  | 1120  | 648   | 238   | 131   | 257   | 638   | 1319  | 2020  | 2513  | (9)  |
| (10) |                                                                   | CDD50       | 867   | 0     | 0     | 0     | 0     | 33    | 232   | 358   | 217   | 27    | 0     | 0     | 0     | (10) |
| (11) |                                                                   | CDD65       | 34    | 0     | 0     | 0     | 0     | 0     | 9     | 24    | 1     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                   | CDH74       | 1022  | 0     | 0     | 0     | 0     | 63    | 356   | 473   | 128   | 2     | 0     | 0     | 0     | (12) |
| (13) |                                                                   | CDH80       | 184   | 0     | 0     | 0     | 0     | 10    | 63    | 98    | 13    | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg       | 2.1   | 1.8   | 2.1   | 2.1   | 2.9   | 3.0   | 2.2   | 1.7   | 1.7   | 2.1   | 2.4   | 1.9   | 1.7   | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 23.0  | 0.2   | 0.3   | 0.5   | 1.5   | 2.0   | 3.7   | 4.9   | 4.4   | 3.3   | 1.1   | 0.7   | 0.4   | (15) |
| (16) |                                                                   | PrecMax     | 34.4  | 0.8   | 0.8   | 1.9   | 3.0   | 5.9   | 7.2   | 9.3   | 10.2  | 7.0   | 3.0   | 2.0   | 1.9   | (16) |
| (17) |                                                                   | PrecMin     | 14.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.5   | 0.9   | 1.7   | 0.9   | 0.3   | 0.1   | 0.1   | 0.0   | (17) |
| (18) |                                                                   | PrecStd     | 4.5   | 0.2   | 0.2   | 0.4   | 0.8   | 1.3   | 1.9   | 2.1   | 2.1   | 1.6   | 0.7   | 0.4   | 0.4   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 16.4  | 22.8  | 43.0  | 58.5  | 80.7  | 85.0  | 86.5  | 81.7  | 72.8  | 55.9  | 26.6  | 15.1  | (19) |
| (20) |                                                                   |             | MCWB  | 13.6  | 18.4  | 33.1  | 43.0  | 57.8  | 63.7  | 65.2  | 64.1  | 56.9  | 43.4  | 22.7  | 12.6  | (20) |
| (21) |                                                                   | 2%          | DB    | 10.8  | 17.8  | 35.0  | 52.3  | 74.5  | 81.5  | 82.6  | 77.4  | 67.6  | 48.6  | 21.2  | 9.1   | (21) |
| (22) |                                                                   |             | MCWB  | 9.0   | 14.9  | 27.9  | 39.6  | 55.1  | 61.7  | 64.4  | 62.0  | 55.2  | 38.0  | 17.9  | 7.4   | (22) |
| (23) |                                                                   | 5%          | DB    | 4.7   | 13.5  | 30.7  | 47.7  | 68.5  | 78.3  | 79.6  | 74.2  | 63.4  | 43.1  | 17.3  | 5.4   | (23) |
| (24) |                                                                   |             | MCWB  | 3.4   | 10.8  | 25.6  | 37.5  | 51.2  | 60.3  | 64.5  | 61.2  | 52.8  | 35.0  | 14.8  | 3.9   | (24) |
| (25) |                                                                   | 10%         | DB    | -0.5  | 9.0   | 26.5  | 43.9  | 62.5  | 74.3  | 75.7  | 70.8  | 58.7  | 38.3  | 13.2  | 0.8   | (25) |
| (26) |                                                                   |             | MCWB  | -1.8  | 6.8   | 22.2  | 34.5  | 47.6  | 59.3  | 63.3  | 59.6  | 50.8  | 32.6  | 11.1  | -0.4  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 13.2  | 18.9  | 33.5  | 44.4  | 59.3  | 68.6  | 69.8  | 67.8  | 61.5  | 43.7  | 22.6  | 12.8  | (27) |
| (28) |                                                                   |             | MCDWB | 15.7  | 21.9  | 41.5  | 54.0  | 76.0  | 77.4  | 78.2  | 74.5  | 66.9  | 54.4  | 26.1  | 14.6  | (28) |
| (29) |                                                                   | 2%          | WB    | 9.1   | 15.0  | 29.1  | 40.5  | 55.5  | 65.3  | 67.9  | 64.9  | 57.7  | 38.7  | 18.1  | 7.5   | (29) |
| (30) |                                                                   |             | MCDWB | 11.1  | 17.8  | 33.8  | 51.8  | 72.8  | 74.4  | 77.6  | 71.7  | 62.1  | 46.9  | 20.9  | 8.9   | (30) |
| (31) |                                                                   | 5%          | WB    | 3.2   | 10.8  | 25.7  | 37.4  | 52.2  | 63.2  | 66.3  | 63.0  | 55.2  | 36.0  | 14.6  | 3.8   | (31) |
| (32) |                                                                   |             | MCDWB | 4.6   | 13.4  | 30.9  | 46.8  | 66.8  | 72.7  | 75.5  | 70.3  | 60.5  | 42.0  | 17.1  | 5.2   | (32) |
| (33) |                                                                   | 10%         | WB    | -1.6  | 6.9   | 21.8  | 35.0  | 48.8  | 61.1  | 64.7  | 61.2  | 52.6  | 33.1  | 11.2  | -0.4  | (33) |
| (34) |                                                                   |             | MCDWB | -0.4  | 8.8   | 26.4  | 42.7  | 60.6  | 71.2  | 73.4  | 68.2  | 58.2  | 38.0  | 13.1  | 0.7   | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 26.0  | 31.6  | 32.9  | 24.4  | 26.6  | 27.4  | 23.8  | 25.6  | 24.6  | 25.3  | 25.8  | 23.3  | (35) |
| (36) |                                                                   |             | MCDBR | 28.0  | 33.1  | 35.6  | 30.7  | 40.7  | 36.1  | 33.1  | 33.7  | 33.9  | 30.5  | 29.8  | 25.3  | (36) |
| (37) |                                                                   |             | MCWBR | 26.2  | 29.6  | 29.1  | 19.6  | 21.7  | 17.9  | 16.6  | 19.4  | 21.2  | 20.6  | 26.3  | 23.7  | (37) |
| (38) |                                                                   | 5% WB       | MCDBR | 27.7  | 32.4  | 32.8  | 28.3  | 38.5  | 27.9  | 25.8  | 27.5  | 24.2  | 29.0  | 29.5  | 25.2  | (38) |
| (39) |                                                                   |             | MCWBR | 25.9  | 29.3  | 26.8  | 18.4  | 21.1  | 15.2  | 14.7  | 18.3  | 16.6  | 19.9  | 26.2  | 23.6  | (39) |
|      |                                                                   |             |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.239 | 0.273 | 0.317 | 0.439 | 0.489 | 0.474 | 0.481 | 0.406 | 0.353 | 0.323 | 0.265 | 0.234 | (40)  |      |
| (41) |                                                                   | taud        | 2.132 | 2.105 | 2.041 | 1.857 | 1.926 | 2.075 | 2.093 | 2.309 | 2.398 | 2.274 | 2.121 | 2.071 | (41)  |      |
| (42) |                                                                   | Ebn at Noon | 222   | 254   | 266   | 244   | 240   | 246   | 242   | 256   | 257   | 232   | 203   | 191   | (42)  |      |
| (43) |                                                                   | Edh at Noon | 19    | 29    | 40    | 56    | 56    | 49    | 47    | 36    | 28    | 25    | 19    | 16    | (43)  |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 344   | 715   | 1290  | 1724  | 1627  | 1826  | 1572  | 1365  | 1012  | 655   | 389   | 247   | (44)  |      |
| (45) |                                                                   | RadStd      | 13    | 21    | 44    | 115   | 121   | 143   | 178   | 128   | 92    | 43    | 24    | 10    | (45)  |      |

## Historical Trends

|                             |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                         | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) Station Only           | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (5 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page